1 Phase Kickback

Let U be a unitary operator with eigenstate $|\psi\rangle$ and eigenvalue $e^{2\pi i\theta}$:

$$U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle.$$

(a) Consider a two-qubit system initialized in state $|+\rangle|\psi\rangle$, where $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$. Suppose we apply the gate U to the qubit in state $|\psi\rangle$ conditioned on the first qubit being in state $|1\rangle$. What is the resulting two-qubit state?

Following the controlled-U gate, the state of the two qubits is given by

$$\begin{aligned}|\Psi\rangle &= \frac{1}{\sqrt{2}}\left(|0\rangle|\psi\rangle + |1\rangle U|\psi\rangle\right)\\ &= \frac{1}{\sqrt{2}}\left(|0\rangle|\psi\rangle + e^{2\pi i\theta}|1\rangle|\psi\rangle\right)\\ &= \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i\theta}|1\rangle\right)|\psi\rangle.\end{aligned}$$

(b) Let $U = Z$ and $|\psi\rangle = |1\rangle$ (with $Z|1\rangle = -|1\rangle$). What is the state of the control qubit after the controlled-Z gate?

Since $U|\psi\rangle = -|\psi\rangle$, we have $\theta = 1/2$. Using our result from part (a), the state of the control qubit following the CZ gate is therefore $(|0\rangle - |1\rangle)/\sqrt{2}$.

2 Binary Fractions

In a QPE circuit with m ancilla qubits, the jth ancilla qubit controls the application of U^{2^j} for $j = 0, \ldots, m-1$. Assume the phase θ, defined as above, can be expressed exactly as an m-bit binary fraction:

$$\theta = 0.\theta_1\theta_2\cdots\theta_m = \sum_{k=1}^{m}\theta_k 2^{-k}, \quad \theta_k \in \{0,1\}.$$

(a) What is the decimal representation of the binary fraction 0.111? What is the binary representation of the decimal fraction 0.625? Hint: Note that $0.625 = 5/8$.

To express the binary fraction 0.111 in decimal representation, we can use the formula given above to expand 0.111 as

$$\begin{aligned}0.111 &= 1\times\frac{1}{2} + 1\times\left(\frac{1}{2}\right)^2 + 1\times\left(\frac{1}{2}\right)^3\\ &= \frac{1}{2}+\frac{1}{4}+\frac{1}{8}\\ &= \frac{7}{8}.\end{aligned}$$

Writing 7/8 as a decimal then gives 0.875.

To write $0.625 = 5/8$ in binary, we first note that

$$\frac{5}{8} = 1\times\frac{1}{2} + 0\times\frac{1}{4} + 1\times\frac{1}{8}.$$

The coefficients above then give us the binary representation: 0.101.

(b) Show that for any integer $j \geq 0$, the phase produced by U^{2^j} acting on $|\psi\rangle$ satisfies

$$e^{2\pi i(2^j\theta)} = e^{2\pi i(0.\theta_{j+1}\cdots\theta_m)}.$$

Hint: Try separating $2^j\theta$ into integer and fractional parts, and don't forget that $e^{2\pi im} = 1$ for any $m \in \mathbb{Z}$.

Following the hint, we consider

$$\begin{aligned} 2^j\theta &= \sum_{k=1}^{m} \theta_k 2^{j-k} \\ &= \sum_{k=1}^{j} \theta_k 2^{j-k} + \sum_{k=j+1}^{m} \theta_k 2^{j-k}. \end{aligned}$$

The summation in the first term above contains only non-negative powers of 2 and is therefore an integer. The second term, meanwhile, can be written as

$$\begin{aligned} \sum_{k=j+1}^{m} \theta_k 2^{j-k} &= \sum_{k=1}^{m-j} \theta_{j+k} 2^{-k} \\ &= 0.\theta_{j+1}\theta_{j+2}\cdots\theta_m. \end{aligned}$$

Since the integer part of $2^j\theta$ simply contributes a phase factor of 1, this then proves the result.

(c) Suppose $\theta = 0.625$. Use your results from (a) and (b) to determine the phase factors applied by U^{2^0}, U^{2^1}, and U^{2^2}.

Since $0.625 = 5/8$, the phase factor applied by $U^{2^0} = U$ is simply $e^{2\pi i(5/8)} = e^{5\pi i/4}$. We know from part (a) that the binary representation of 0.625 is 0.101. To find the phase factors applied by U^2 and U^4, we can then use the result of part (b): For U^2, the phase factor is $e^{2\pi i(0.01)} = e^{2\pi i(1/4)} = i$, while for U^4, it is $e^{2\pi i(0.1)} = e^{2\pi i(1/2)} = -1$.

3 The Quantum Fourier Transform

Following the application of the controlled-U gates, the state $|\Psi\rangle$ of all qubits *prior* to the inverse Fourier transform $\mathrm{QFT}^\dagger$ is given by

$$|\Psi\rangle = |\psi\rangle \otimes \frac{1}{\sqrt{2^m}} \sum_{k=0}^{2^m-1} e^{2\pi ik\theta} |k\rangle .$$

In the case where θ is exactly equal to an m-bit binary fraction, we have $\theta = j/2^m$. The integer j can then be recovered as

$$|\psi\rangle \otimes |j\rangle = (\mathbb{1} \otimes \mathrm{QFT}^\dagger) |\Psi\rangle .$$

(a) Consider the case where $\theta \neq j/2^m$. Write the action of $(\mathbb{1} \otimes \mathrm{QFT}^\dagger)$ on $|\Psi\rangle$. This can be left as a sum for now.

Recall that the action of $\mathrm{QFT}^\dagger$ on a computational basis state $|k\rangle$ can be written as

$$\mathrm{QFT}^\dagger |k\rangle = \frac{1}{\sqrt{2^m}} \sum_{n=0}^{2^m-1} e^{-2\pi i n k/2^m} |n\rangle .$$

Applied to the state $|\Psi\rangle$, this then gives

$$(\mathbb{1} \otimes \mathrm{QFT}^\dagger) |\Psi\rangle = |\psi\rangle \otimes \sum_{n=0}^{2^m-1} A_n |n\rangle ,$$

where

$$A_n = \frac{1}{2^m} \sum_{k=0}^{2^m-1} e^{2\pi i k(\theta - n/2^m)}.$$

(b) Using your result from (a), explain why a measurement of the ancilla register now leads (for $\theta \neq j/2^m$) to a probability distribution over computational basis states. Hint: Express $\mathrm{QFT}^\dagger |\Psi\rangle$ so that the coefficients of the computational basis states $\{|j'\rangle\}$ take the form $\sum_{k=0}^{M-1} r^k$ for some r and M, and then apply the following result for summing geometric series:

$$\sum_{k=0}^{M-1} r^k = \frac{1 - r^M}{1 - r}.$$

We start by writing the phase θ as $\theta = (j + \delta)/2^m$, where $\delta = 0$ only when θ is exactly equal to an m-bit binary fraction. We then consider the coefficient A_n from part (a). This coefficient is a geometric series of the form

$$A_n = \frac{1}{2^m} \sum_{k=0}^{2^m-1} r^k,$$

where $r = e^{2\pi i(j+\delta-n)/2^m}$. In the simple case where $\theta = j/2^m$ ($\delta = 0$), then $r = 1$ for $j = n$, and we have

$$A_j = \frac{1}{2^m} \sum_{k=0}^{2^m-1} 1 = 1.$$

This indicates that the probability $P_j = |A_j|^2$ of measuring the outcome $|j\rangle$ is equal to one.

For $\delta \neq 0$, we can use the hint provided to write the summation in closed form as

$$A_n = \frac{1}{2^m} \frac{1 - e^{2\pi i \delta}}{1 - e^{2\pi i(j+\delta-n)/2^m}},$$

where we have simplified the numerator using the fact that $j - n$ is necessarily an integer, and consequently, that $e^{2\pi i(j-n)} = 1$. The probability P_n of obtaining the measurement outcome $|n\rangle$ is then given by

$$P_n = \frac{1}{2^{2m}} \frac{\sin^2(\pi\delta)}{\sin^2\left(\pi \frac{j+\delta-n}{2^m}\right)}.$$

To understand the probability leakage, note that for states n close to j, the difference $|j - n|$ is much less than 2^m. We can then understand the leading qualitative behaviour in P_n by applying the small-angle approximation $\sin^2(x) \approx x^2$ to the denominator, valid for $x \ll 1$. This gives

$$P_n \approx \frac{\sin^2(\pi\delta)}{\pi^2(\delta - \Delta)^2},$$

where $\Delta = n - j$. The probability reaches a maximum value when $\Delta = 0$, but remains non-zero (falling off quadratically) for states with $\Delta \neq 0$.

4 Single-ancilla QPE

Standard QPE assumes we have m ancilla qubits, all of which are measured at the end of the circuit. In this exercise, we consider an alternative setup where we instead have a single ancilla qubit. This ancilla qubit will be measured m times, with a reset in between measurements, and will return a binary digit θ_k of the unknown phase at each measurement.

(a) Assume $U|\psi\rangle = e^{2\pi i\theta}|\psi\rangle$, where $\theta = 0.\theta_1\theta_2\cdots\theta_m$ is an exact m-bit binary fraction. In the first step ($k = m$), the ancilla is initialized to $|+\rangle$, and a controlled $U^{2^{m-1}}$ gate is applied to the target in state $|\psi\rangle$. Write the joint state of the ancilla and target qubits following this gate.

The state $|\Psi\rangle$ of the two qubits following this gate is given by

$$\begin{aligned}|\Psi\rangle &= \frac{1}{\sqrt{2}}\left(|0\rangle|\psi\rangle + |1\rangle U^{2^{m-1}}|\psi\rangle\right) \\ &= \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i(2^{m-1}\theta)}|1\rangle\right)|\psi\rangle\,.\end{aligned}$$

(b) Starting from your result for (a), consider a measurement of the ancilla qubit in the X basis. (Equivalently, you can think about first applying a Hadamard and then measuring in the Z basis.) Show that the measurement outcome $x_m \in \{0, 1\}$ deterministically recovers the least significant bit θ_m of the unknown phase. Hint: This will use the result you proved in Question 2(b).

In a previous question, we proved that

$$2^j\theta = n + \sum_{k=1}^{m-j}\theta_{j+k}2^{-k},$$

where n is an integer. In this case, we have $j = m - 1$, giving

$$\begin{aligned}2^{m-1}\theta &= n + \sum_{k=1}^{1}\theta_{m-1+k}2^{-k} \\ &= n + \theta_m 2^{-1}.\end{aligned}$$

Applying this simplification to the state $|\Psi\rangle$ derived in (a), we then find that

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{\pi i\theta_m}|1\rangle\right)|\psi\rangle\,.$$

If $\theta_m = 0$, then the control qubit is in state $|+\rangle$, whereas for $\theta_m = 1$, it is instead in state $|-\rangle$. A measurement of the control qubit in the $|\pm\rangle$ basis therefore lets us infer the bit value of θ_m.

(c) We now consider the following steps $k = m - 1, m - 2, \ldots, 1$. In each step, assume that the bits $\theta_{k+1}, \cdots, \theta_m$ are known, and that the ancilla has been re-initialized to $|+\rangle$ following the measurement in the preceding step. For step k, the gate applied is a controlled $U^{2^{k-1}}$ gate. We can deterministically extract θ_k from an X-basis measurement of the ancilla, provided we first apply a phase rotation gate $R_z(-\phi_k)$ to the ancilla. Show that the required rotation angle is given by

$$\phi_k = 2\pi\sum_{\ell=k+1}^{m}\theta_\ell 2^{-(\ell-k+1)}.$$

In step k, the state $|\Psi\rangle$ of the ancilla and target qubits following the controlled gate is instead given by

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{2\pi i(2^{k-1}\theta)}|1\rangle\right)|\psi\rangle\,.$$

Using the result shown in 2(b), we can divide $2^{k-1}\theta$ into integer and fractional parts according to

$$\begin{aligned} 2^{k-1}\theta &= n + 0.\theta_k\theta_{k+1}\cdots\theta_m \\ &= n + \frac{\theta_k}{2} + \sum_{\ell=k+1}^{m}\theta_\ell 2^{-(\ell-k+1)}, \end{aligned}$$

where, as before, $n \in \mathbb{Z}$. This then lets us write $|\Psi\rangle$ as

$$|\Psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + e^{\pi i\theta_k}e^{i\phi_k}|1\rangle\right)|\psi\rangle\,,$$

where $\phi_k = 2\pi\sum_{\ell=k+1}^{m}\theta_\ell 2^{-(\ell-k+1)}$. Given knowledge of $\theta_{k+1},\ldots,\theta_m$, this phase can be deterministically compensated through application of a phase gate $R_z(-\phi_k)$. The bit value of θ_k can then be determined as in part (a).